



Submission Title: Automated Generation of Angle and Height Measures in Distal Radius Fracture Recovery Prediction

SUBMISSION PREVIEW: AUTOMATED GENERATION OF ANGLE AND HEIGHT MEASURES IN DISTAL RADIUS FRACTURE RECOVERY PREDICTION

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Abstract

Track/Topics

Biomedical Imaging and Instrumentation

Sub-track:

Deep Learning and Artificial Intelligence In Biomedical Imaging

Introduction:

As part of a larger effort to use machine learning to predict the need for resurgery following reduction of distal radius fracture [1], multiple pre- and post-reduction measures of the relative position and orientation of the radius, ulna, and wrist bones (carpals) have been found to be of high predictive value. The most important of these measures include radial height, ulnar variance, and radial inclination. Typically, such anatomical quantities are obtained through manual computer-based measurement performed by experienced radiologists or orthopaedic surgeons. This requires extensive time and can involve a degree of human error, as inter-rater reliability is not 100%. Therefore, it would be desirable, and more robust, to have a fully-automated means to obtain such measures from collected radiographs. As a first step toward this goal, this project demonstrates how state-of-the-art segmentation models can be used to automatically generate accurate measurements for many of the desired critical criteria.

Materials and Methods:

DATA: For this project, a set of wrist radiographs was obtained from a range of public internet sources (e.g., Radiopaedia [2]).

ALGORITHM: Our pilot measurement process operates as follows. First, an automated segmentation model (Sam-med2D [3]) obtains the outline of each of the wrist bones (e.g., radius, ulna, carpals), thereby providing spatial masks for the bones of interest. Second, the matching squares algorithm from scikit-image [4] is used to create a smooth, continuous outline (boundary) for each of the spatial masks generated previously. These boundaries are then used to calculate values for each of the desired clinical measurements, based on established definitions [5,6].

MEASUREMENTS: For this pilot, automated calculation of ulnar variance, radial inclination, and radial height was achieved. These three measures are defined and obtained as follows. (1) Ulnar variance is the difference between the length of the distal articular surface of the ulna and that of the radius, obtained following automated detection of the width of each indicated surface. (2) Radial inclination is the slope of the articular surface relative to the long axis of the bone, obtained as the angle between a line perpendicular to the radius and a line drawn along the superior surface, from the radial styloid to the ulnar border. (3) Radial height is the height of the radial styloid, obtained as the vertical difference (in an AP radiograph) from the top of the radial styloid to the superior-most point of the ulnar aspect of the articular surface.

Results, Conclusions, and Discussions:

Figure 1 depicts, on the normal left wrist radiograph of [2], the result of automated generation of masks for the radius (green), ulna (blue), and a carpal (magenta), along with the automated generation of their associated boundaries (black line).

Figure 2 illustrates sample measures for each of the three quantities identified above. Note that in the absence of a known scale, the ulnar variance and radial height are given in pixel counts. In a clinical setting, the use of a known scale reference would permit direct calculation of these quantities in the desired units of length.

Having generated quantities shown in [1] to be of high predictive value, we will use the distal radius fracture data from [1] to obtain multiple human rater (orthopaedic surgeons and fellows) measures for each case. We will then evaluate the performance of our automated algorithm against the expert raters to confirm whether it may serve as a more rapid and robust means to obtain quantities that have high predictive value for post-reduction risk of resurgery.

Acknowledgements and/or References (Optional):

- [1] Lyu, R., Taleghani, E., Hale, I., Schackelford, T., Rex, J., Florczynski, M., & Talavage, T. (2025) "Machine Learning Prediction of Radiographic Outcomes for Nonoperatively Treated Distal Radius Fractures," BMES 2025.
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- [6] Blakeney, W. G. (2010). Stabilization and treatment of Colles' fractures in elderly patients. Clinical interventions in aging, 337-344.

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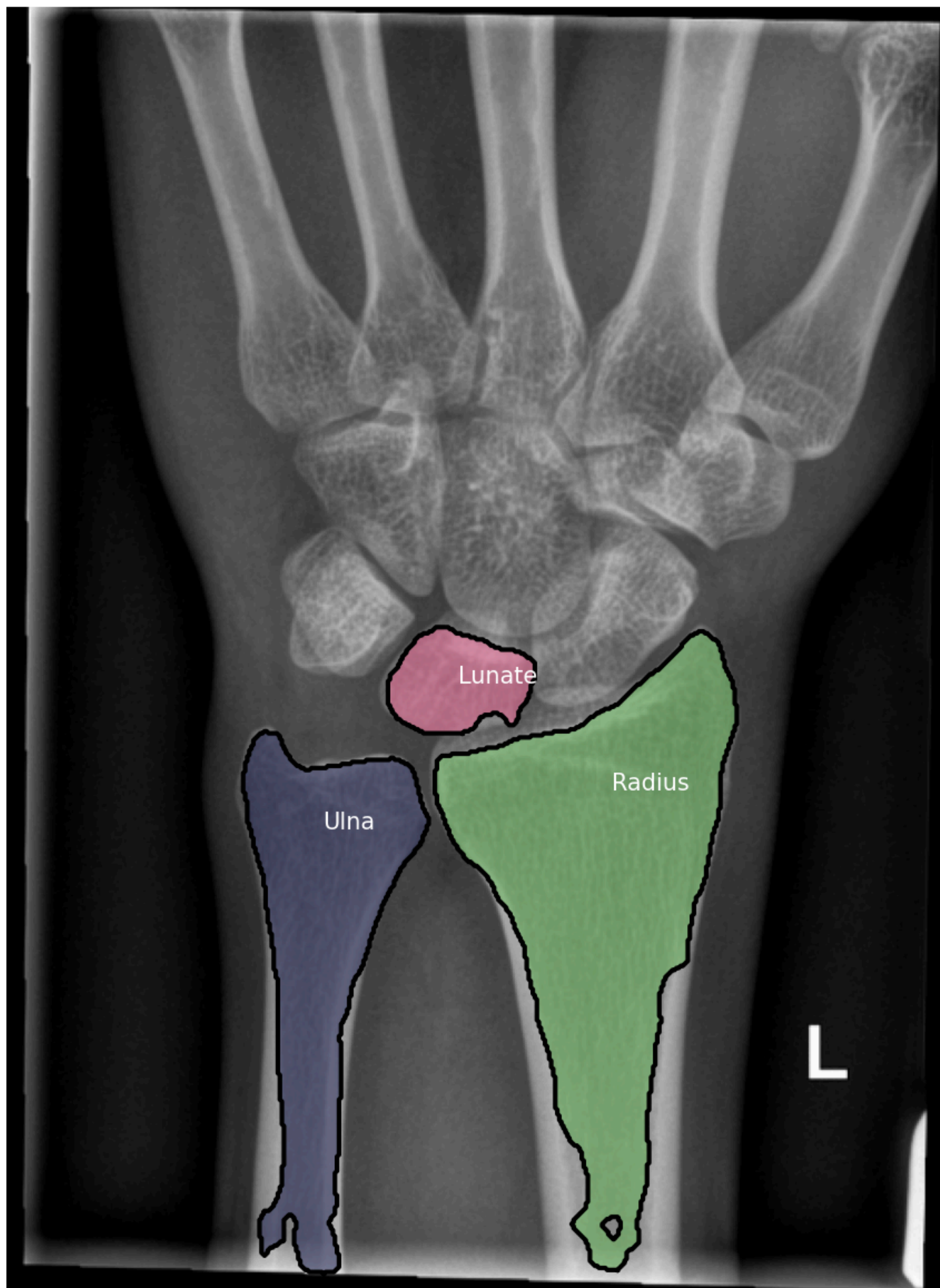


Figure 1: Example AP radiograph on which Sam-med2D has been used to obtain masks for each of the radius (green; right), ulna (blue; left) and the lunate bone (magenta; top), following which scikit-image has been used to generate continuous outlines (black lines). AP radiograph obtained from [2].

New_Ozer_Figure1.png

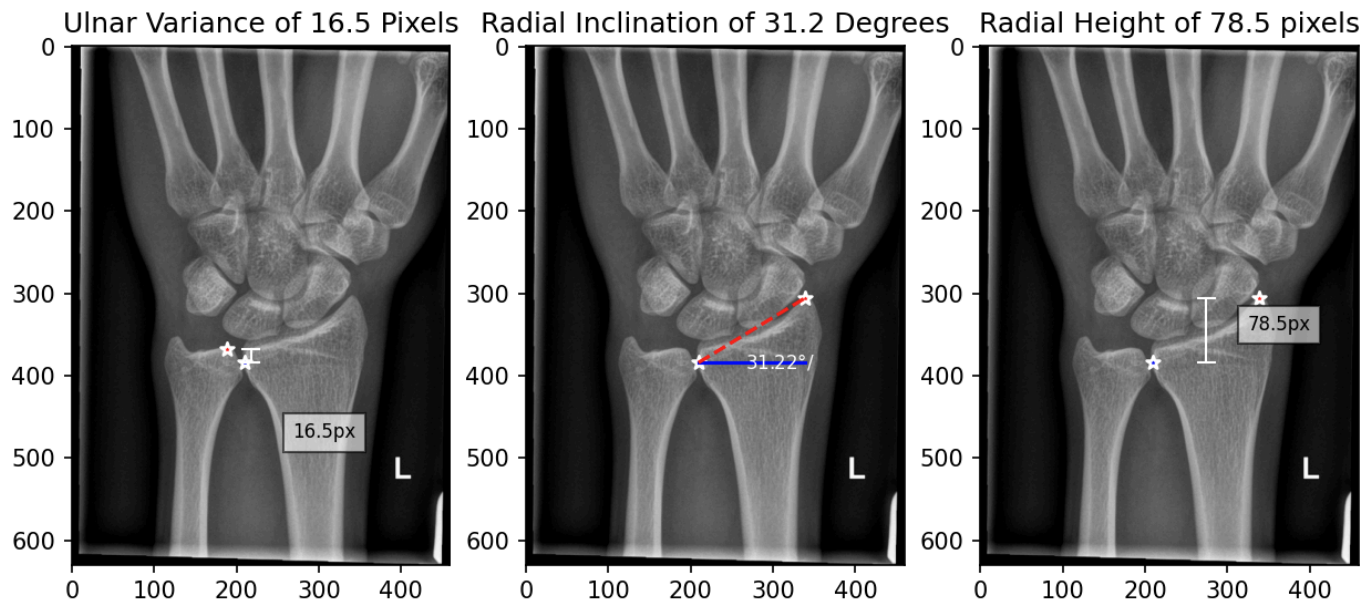


Figure 2: Example of the use of the AP radiograph from Figure 1 (from [2]) for automated computation of (left) ulnar variance, (middle) radial inclination, and (right) radial height. Note that distance measurements are in pixels due to the absence of a standardized reference in the images.

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